

# A Packing of 15 Four-Spaces in $\mathbb{R}^{11}$

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## Abstract

We give 15 4-dimensional subspaces of  $\mathbb{R}^{11}$  whose smallest squared chordal distance is at least 2.726912. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.714783, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 15 subspaces of dimension 4 in  $\mathbb{R}^{11}$  whose pairwise squared chordal distances  $D(a, b) = 4 - \text{tr}(P_a P_b)$  are all at least*

$$D_{\min} = 2.726912 \dots$$

The exact value of  $D_{\min}$ , a rational number, is given in Appendix A.

## 2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some  $N$  in the row  $m = 11$ ,  $n = 4$ . The cell  $N = 15$  is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.714783, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is  $4(11 - 4)/11 \cdot 15/14 = 2.727273$ , and the value above lies below it.

## 3 Verification

Each subspace is given by an integer  $m \times n$  basis  $Y$  (coordinates times  $10^{12}$ , rounded). With  $G = Y^\top Y$  the projection is  $P = YG^{-1}Y^\top$ , so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N15.txt`

in <https://github.com/Ge-limin/math719>.

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\*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

## 4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

## A Exact value

numerator:

```
272691206714907147904884367521103838592886196464121667786508328889736224
795397219166435885944753176646383336637935476145664026372888292350093148
7635333292322242603485561848935863688551819756699
```

denominator:

```
999999999999500281125240998756587409982850542687208109663603549577997443
813597577377531760577960485754286936830066113281465620375161536889234637
712665903299401600731701166172079685901251648291
```

## B Coordinates

15 blocks of 11 rows with 4 integers each; divide by  $10^{12}$ . File: `grassmann/packings/m11n4N15.txt`.

```
-410226257312 60716877626 350939817160 18354649161401397896304 -79085483086 405069159030 -6743854518635460923575 -176828349393 -282752930843 211499167436
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